

Problem Statement

Preparing for coding interviews often requires solving hundreds of LeetCode problems. While solutions and explanations are available online, candidates frequently spend significant time searching for problem details, understanding approaches, analyzing complexities, and creating study notes.

The process is repetitive and time-consuming, especially when revisiting previously solved problems for revision.

To address this challenge, an automated solution is needed that can:

- Accept a LeetCode problem number as input.
- Automatically identify the corresponding LeetCode problem.
- Generate structured study notes including:
 - Problem Statement
 - ELI10 Explanation
 - Intuition
 - Brute Force Approach
 - Optimal Approach
 - Time Complexity Analysis
 - Space Complexity Analysis
 - Python Solution
 - Dry Run Example
- Export the generated content into a Markdown file for future reference.

The objective of this project is to build an AI-powered LeetCode learning assistant using n8n, SERPER API, and OpenRouter that automates problem discovery and note generation, helping students learn and revise Data Structures and Algorithms more efficiently.